
Time Series Analysis in Network Traffic: Anomaly Detection for Cybersecurity

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Abstract

Accurate forecasting of network traffic is a crucial component of modern cybersecurity, enabling Internet Service Providers (ISPs) to detect irregularities before they result in critical service interruptions. This thesis evaluates the efficacy of Deep Learning methods for unsupervised anomaly identification in large-scale academic networks using the CESNET-TimeSeries24 dataset. By examining the effects of data preprocessing, architectural complexity, and training strategies, we explore the trade-offs between model generalization and operational overhead.

Our findings challenge the assumption that highly specialized or deep architectures are required for heterogeneous network traffic. We demonstrate that a lightweight, 1-Layer Gated Recurrent Unit (GRU) utilizing a global training strategy outperforms specialized clustering approaches, achieving a 98.5% win rate and zero catastrophic failures across 541 distinct subnets. Furthermore, we establish that log-arithmetic scaling is mandatory for stabilizing gradient descent against the extreme heavy-tailed distribution of network traffic. Building upon this forecasting engine, we develop a dynamic, reconstruction-based anomaly detection framework. Case studies demonstrate the system's capability to effectively identify both volumetric security incidents (point anomalies) and protocol-level infrastructure breakdowns (contextual anomalies), providing a robust and scalable baseline for real-time ISP monitoring.

1 Introduction

1.1 Context and Motivation

The rapid expansion of digital services and high-bandwidth applications has led to an exponential increase in global network traffic. For Internet Service Providers (ISPs) and National Research and Education Networks (NRENs), such as CESNET, maintaining infrastructure stability is critical. Network administrators face the dual challenge of resource optimization and security assurance. Specifically, the ability to distinguish between legitimate high-volume data transfers (e.g., academic backups) and malicious activities (e.g., DDoS attacks, data exfiltration) is a fundamental requirement for modern network defense. Traditional monitoring approaches rely heavily on static thresholding (e.g., alerting if traffic exceeds a fixed Mbps limit) or simple statistical methods. However, these techniques are increasingly ineffective in modern environments where traffic patterns are highly stochastic, non-linear, and subject to complex seasonal dependencies.

1.2 Problem Statement

The failure of static methods necessitates the adoption of advanced Machine Learning (ML) techniques. Although deep learning, specifically Recurrent Neural Networks (RNNs), has shown promise in time-series forecasting, its application to network traffic remains challenging due to three factors:

1. **Computational Overhead:** Deep models (e.g., multi-layer LSTMs) are often computationally expensive, raising questions about their feasibility for real-time monitoring of thousands of subnets.
2. **Heterogeneity:** Network subnets exhibit diverse behaviors—some strictly periodic, while others chaotic. It is unclear if a single "Global" model can generalize across this diversity or if complex clustering strategies are required.
3. **Anomaly Definition:** Distinguishing between "contextual anomalies" (e.g., service outages) and simple volume spikes requires a system capable of understanding temporal context, not just magnitude.

1.3 Research Objectives

To address these challenges, this thesis investigates the optimization and application of Deep Learning models for anomaly detection in the CESNET network. The research is guided by the following four questions:

- **RQ1: Architecture Efficiency & Optimization**
How does the complexity of state-of-the-art deep learning models (e.g., GRU vs. LSTM, 1-Layer vs. 3-Layers) impact the trade-off between forecasting accuracy and computational efficiency in network traffic?
- **RQ2: Training Strategy**
Does grouping subnets based on behavioral clustering improve forecasting performance compared to a unified Global model?
- **RQ3: Data Aggregation & Predictability**
How does the granularity of traffic aggregation (IP addresses vs. Subnet vs. Institution) influence the ability of deep learning models to capture temporal dynamics?
- **RQ4: Applicability to Anomaly Detection**
Can reconstruction-based deep learning models with dynamic thresholding effectively detect contextual anomalies and outages in real-world network traffic?

1.4 Thesis Contributions

This study contributes to the field of network traffic analysis by establishing a reproducible framework for real-time anomaly detection. The key contributions are:

1. **Architectural Optimization:** We demonstrate that a lightweight **1-Layer GRU** architecture matches the accuracy of deeper, complex models while significantly reducing training time.
2. **Evaluation of Training Strategies:** We provide empirical evidence that a unified Global model generalizes effectively across diverse subnets, rendering complex clustering strategies unnecessary for this domain.
3. **Forensic Diagnosis Framework:** We implement an unsupervised detection system utilizing a dynamic threshold (3σ). Through case studies, we demonstrate the system's ability to detect widely differing anomaly types, including volumetric attacks (Subnet 1) and infrastructure outages (Subnet 9).

2 Dataset

Before defining the forecasting methodology, we conducted a comprehensive statistical analysis of the *CESNET-TimeSeries24* dataset. This exploratory phase was critical to understand the underlying statistical properties of network traffic and to inform feature selection.

2.1 Dataset Overview

The dataset consists of real-world network telemetry collected from the CESNET (Czech Education and Science Network) backbone.

- **Scope:** It covers 40 weeks of traffic data across **283 institutions, 548 distinct institutional subnets and over 270,000 individual IP addresses.**, representing a diverse mix of academic institutions, dormitories, and administrative offices.
- **Granularity:** The data is aggregated at three distinct intervals: **10 minutes, 1 hour, 1 day**. Each interval-specific time series contains multiple metrics useful for network traffic forecasting and anomaly detection, as listed in Table 1. Given that the original capture interval was 10 minutes, metrics that represent unique values were adapted for the 1 hour and 1 day series incorporating sum, average and standard deviation statistics to enhance temporal granularity.
- **Volume:** The dataset captures the full spectrum of network activity, from idle subnets transferring a few kilobytes to backbone links handling petabytes of data.

Spanning a substantial 40-week period, from October 2023 to July 2024, the dataset captures both long-term trends and fine-grained fluctuations, offering an invaluable resource for rigorous deep-learning model assessment. This dataset enables a detailed examination of the model’s scalability, adaptability, and performance in forecasting diverse traffic patterns within a real-world ISP environment.

Table 1: Time series metrics available in the CESNET dataset.

Metric Name	Description
n_flows	The number of IP flows
n_packets	The number of packets
n_bytes	The number of bytes
n_dest_ip	The number of unique destination IPs
n_dest_ports	The number of unique destination ports
n_dest_asn	The number of unique destination ASNs
tcp_udp_ratio_packets	The TCP/UDP packet ratio
tcp_udp_ratio_bytes	The TCP/UDP bytes ratio
dir_ratio_packets	The directional ratio of packets
dir_ratio_bytes	The directional ratio of bytes
avg_duration	The average duration
avg_ttl	The average TTL

2.2 Feature Selection and Dimensionality Reduction

To minimize computational complexity and avoid multicollinearity, we conducted a statistical analysis on the raw network traffic features to identify redundant metrics.

2.2.1 Correlation Analysis

We examined the correlation coefficient (r) between the three primary volume metrics: `n_bytes`, `n_packets`, and `n_flows`. As shown in Figure 1b, the analysis revealed a significant collinearity between byte volume and packet counts:

- **Bytes vs. Packets:** $r = 0.9189$ (Extremely Strong Correlation)
- **Packets vs. Flows:** $r = 0.77$ (Strong Correlation)
- **Bytes vs. Flows:** $r = 0.48$ (Moderate Correlation)

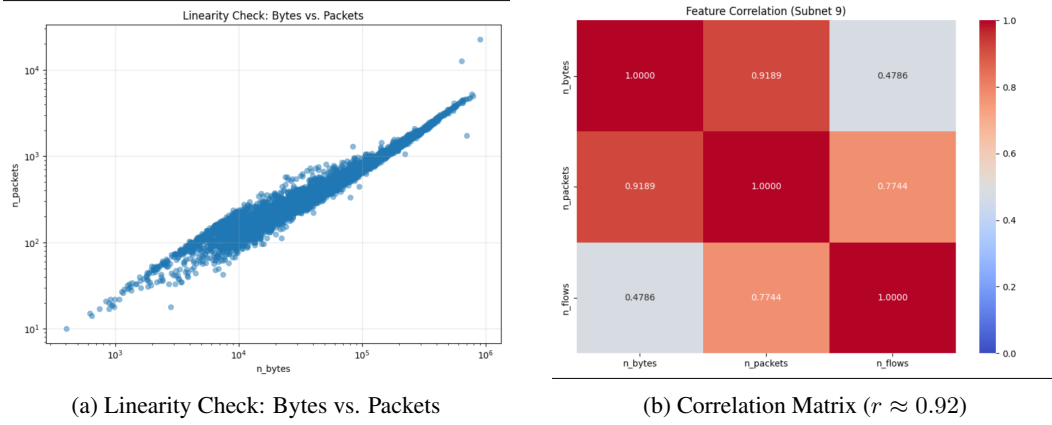


Figure 1: Feature Analysis: (a) Visual confirmation of the tight linear relationship between bytes and packets across logarithmic scales. (b) Statistical confirmation showing extremely high correlation (0.92), justifying the removal of redundant features.

2.2.2 Justification for n_bytes Selection

Based on these findings, we selected n_bytes as the sole univariate feature for model training. This decision is justified by three factors:

1. **Redundancy Elimination:** The correlation of ≈ 0.92 indicates that $n_packets$ provides negligible unique information compared to n_bytes . Including both would likely induce model overfitting due to multicollinearity without adding predictive value.
2. **Infrastructure Relevance:** For Internet Service Providers (ISPs), link saturation is defined by bandwidth capacity (bits per second). Therefore, anomalies in byte volume are the most direct indicators of infrastructure stress (e.g., congestion or DDoS floods).
3. **Sensitivity to "Heavy Hitters":** Unlike flow counts, byte volume captures large data transfer events (e.g., massive backups). A single "Heavy Hitter" flow might look normal in terms of n_flows (count = 1) but will register as a significant spike in n_bytes .

2.3 Statistical Distribution and Scale

A defining characteristic of the CESNET dataset, common to Internet traffic, is its "Heavy-Tailed" distribution. The traffic volume does not follow a normal Gaussian distribution; instead, it is highly skewed.

- **Dynamic Range:** The data spans several orders of magnitude, ranging from idle subnets transferring mere kilobytes (10^3) to backbone links handling petabytes (10^{15}).
- **The "Elephant Flow" Phenomenon:** A small fraction of time steps account for the vast majority of total volume.

Implication for Deep Learning:

As illustrated in Figure 2, training neural networks on such raw, unscaled data causes gradient instability. The logarithmic transformation ($\log(x + 1)$) compresses this dynamic range into a Gaussian distribution, creating a stable optimization landscape.

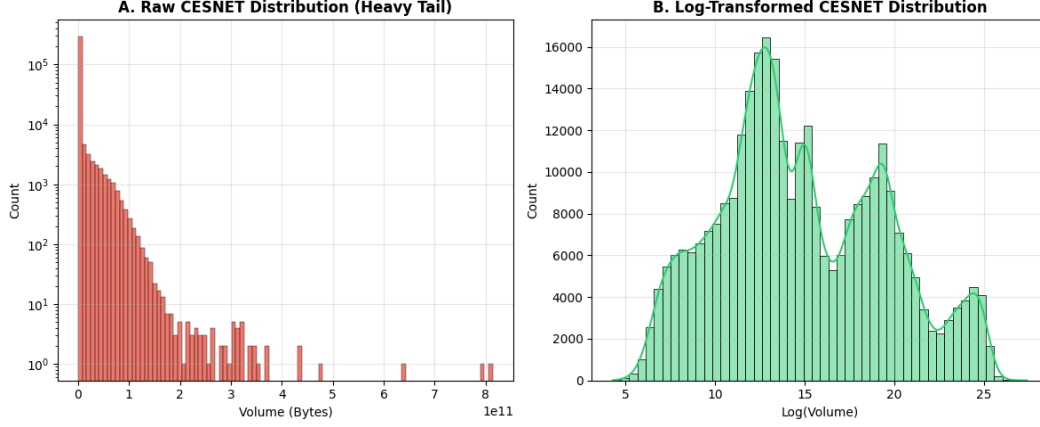


Figure 2: Impact of Preprocessing: (Left) The raw traffic distribution is highly skewed, dominated by low values with extreme outliers. (Right) Logarithmic scaling transforms the data into a normalized distribution suitable for Deep Learning optimization.

2.4 Temporal Dynamics and Heterogeneity

A critical challenge in this dataset is the behavioral heterogeneity across the 548 subnets. Visual inspection reveals two distinct traffic archetypes:

1. **Periodic Subnets (e.g., Subnet 9):** These subnets exhibit strong diurnal (24-hour) and weekly (168-hour) seasonality, driven by regular human working schedules. As shown in Figure 3(a), the traffic follows a clean sine-wave-like pattern, making it highly predictable.
2. **Stochastic Subnets (e.g., Subnet 28):** These subnets are characterized by high volatility and "burstiness," likely due to automated processes or irregular usage. As shown in Figure 3(b), the signal is dominated by noise with weak seasonality, presenting a significant generalization challenge for the forecasting model.

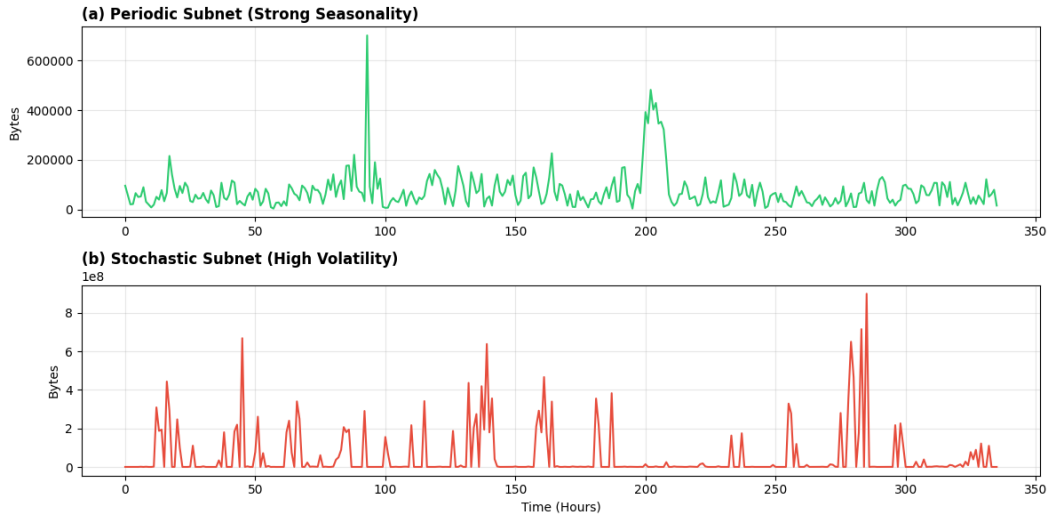


Figure 3: Behavioral Heterogeneity: (a) Subnet 9 shows a clear, rhythmic daily pattern (Periodic). (b) Subnet 28 shows erratic, bursty behavior (Stochastic). A robust Global model must learn to generalize across both types.

3 Methodology

This chapter details the data processing pipeline implemented to transform raw telemetry into tensor-compatible sequences for deep learning. The pipeline, implemented using the *Darts* library, consists of five sequential stages.

3.1 Data Preprocessing Pipeline

3.1.1 Time Alignment and Synchronization

Raw subnet captures often contain irregular timestamps or gaps due to probe downtime. To ensure temporal consistency, we first established a master time index ('times_df') spanning the entire 40-week observation period at strict 1-hour intervals.

- **Merging:** Each subnet's raw data was merged onto this master index using a Left Join operation.
- **Result:** This guarantees that every time series has identical length and aligned timestamps, a prerequisite for batched training.

3.1.2 Missing Value Imputation

Following synchronization, missing entries (NaNs) created by network silence or alignment gaps were handled using the `MissingValuesFiller`. We applied linear interpolation to approximate missing values based on adjacent valid points, ensuring signal continuity without introducing artificial zeros that could distort the distribution.

3.1.3 Chronological Evaluation Splitting

To ensure comparability with prior literature and benchmarks established for the CESNET dataset [Koumar et al. \[2024\]](#), we adopted a highly aggressive chronological partitioning protocol. Unlike standard deep learning splits (e.g., 80/20), we prioritized long-term evaluation:

- **Training Set (First 35% $\approx$ 14 Weeks):** Used to learn the fundamental traffic patterns and seasonality.
- **Validation Set (Next 5% $\approx$ 2 Weeks):** Used for hyperparameter tuning and early stopping.
- **Test Set (Remaining 60% $\approx$ 24 Weeks):** A significantly extended horizon reserved for evaluating generalization.

Justification for Asymmetric Split: This partitioning strategy serves as a rigorous "Stress Test" for two reasons:

1. **Generalization vs. Memorization:** By testing on a horizon nearly double the length of the training period, we verify that the model has learned robust temporal dynamics rather than merely memorizing recent history.
2. **Operational Validity:** Unlike iterative statistical methods (e.g., ARIMA) that often require frequent retraining, our models were trained once and evaluated on the entire 24-week test set without update. Success on this split demonstrates the model's stability and suitability for low-maintenance, real-time deployment.

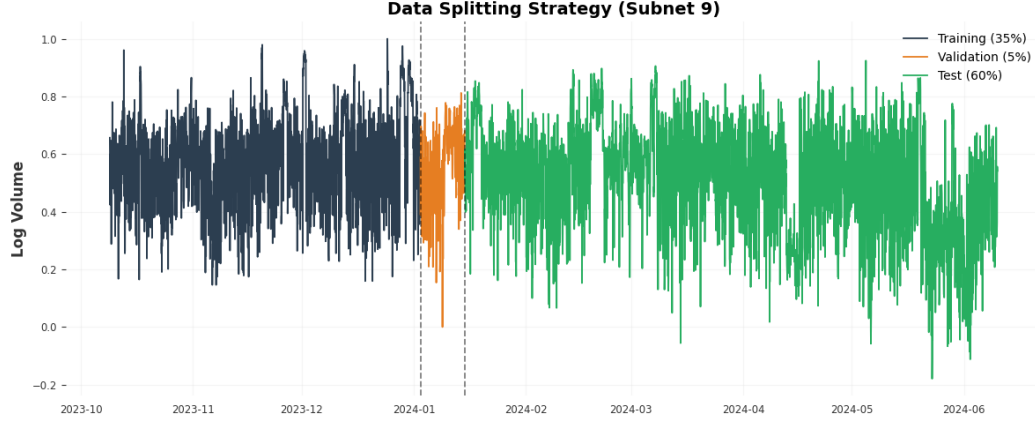


Figure 4: Chronological Data Splitting: The model is trained on only the first 14 weeks (Blue) but must generalize to predict the subsequent 24 weeks (Green). This tests the model’s long-term robustness without retraining.

3.2 Statistical Baselines

To validate the necessity of Deep Learning, we established a performance floor using low-complexity statistical methods. These baselines serve as a reference point; any advanced model must significantly outperform these metrics to justify its computational cost.

3.2.1 Baseline Algorithms

We implemented two non-parametric heuristics evaluated on a randomly selected subset of **30 subnets** using the identical 35/5/60 chronological split described in Section 3.1.3.

A. Historical Mean Model: This model calculates the global average of the training set and projects it as a constant forecast for the entire test horizon.

$$\hat{y}_{test} = \frac{1}{N_{train}} \sum_{t=1}^{N_{train}} y_t \quad (1)$$

While simplistic, this baseline is effective for stationary data. However, as shown in Figure 5a, it fails to capture the seasonality of network traffic, resulting in a "flat-line" prediction.

B. Simple Exponential Smoothing (SES): We utilized the Holt-Winters Exponential Smoothing algorithm (configured with `trend=None`, `seasonal=None`) to model the level component of the series.

$$\hat{y}_{t+1} = \alpha y_t + (1 - \alpha) \hat{y}_t \quad (2)$$

SES weighs recent observations more heavily than older ones. However, without explicit seasonal components, it also converges to a flat forecast for long-term horizons (Figure 5b).

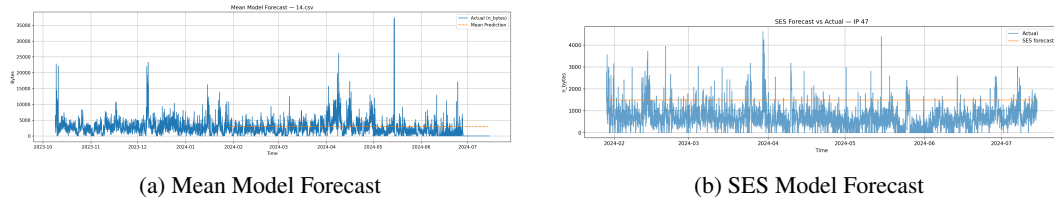
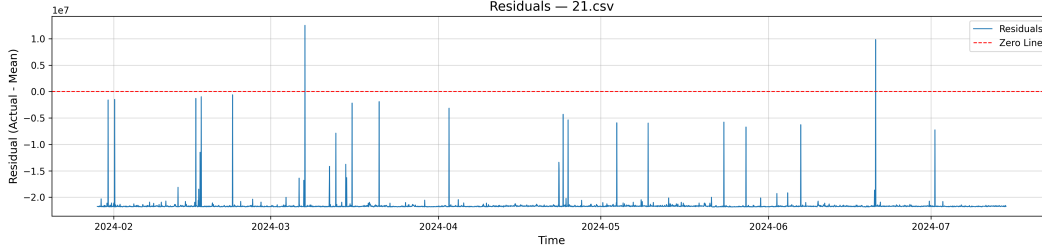


Figure 5: Baseline Performance: Both the Mean (a) and SES (b) models produce flat-line forecasts that completely miss the daily and weekly traffic patterns, justifying the need for Recurrent Neural Networks.

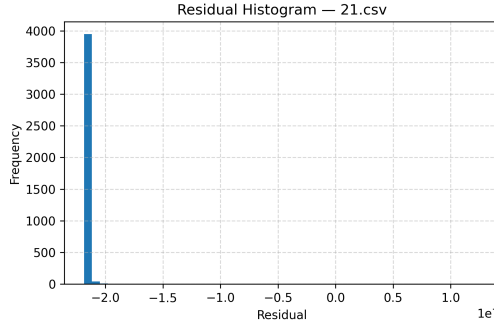
3.2.2 Residual Analysis & Interpretation

We analyzed the residuals ($e_t = y_t - \hat{y}_t$) to diagnose the failure modes of the statistical baselines.

1. **Structured Error (Seasonality):** As illustrated in Figure 6(a), the residuals are not random white noise. Instead, they retain the distinct diurnal and weekly periodicity of the original traffic. This indicates that the baseline models failed to capture any temporal dependencies.
2. **Distribution Shift (Concept Drift):** The histogram in Figure 6(b) reveals a critical failure: the residuals are not centered at zero but are heavily skewed towards negative values (≈ -2.5).
 - **Interpretation:** Since $Residual = Actual - Predicted$, a negative mean indicates that the actual traffic in the test set was consistently lower than the historical average calculated during training.
 - **Conclusion:** This confirms that the network traffic is *non-stationary* (i.e., the mean changes over time due to semester breaks or seasonal shifts). A static Mean Model cannot adapt to these long-term trends, resulting in a systematic bias.



(a) Residuals over Time



(b) Residual Distribution

Figure 6: Error Analysis: The residuals retain the seasonal structure (a), proving the model "under-fitted" the data. The histogram (b) is skewed, violating statistical assumptions of normality.

Table 2: Performance of Statistical Baselines on Representative Subnets. The consistently negative R^2 scores indicate that both the Mean and ETS models fail to capture the variance in the data, often performing worse than a simple random guess due to concept drift.

Subnet ID	Historical Mean		ETS (Additive)	
	RMSE	R^2 Score	RMSE	R^2 Score
Subnet 10	3.13×10^8	-0.064	3.39×10^8	-0.248
Subnet 11	3.85×10^7	-0.001	3.95×10^7	-0.055
Subnet 12	9.04×10^8	-0.041	9.76×10^8	-0.214

Analysis of Table 2: The empirical results reveal a critical limitation of linear statistical methods in this domain:

1. **Negative R^2 :** Across all tested subnets, the Coefficient of Determination (R^2) is negative. This implies that the residual sum of squares is larger than the total sum of squares—in other words, the models are less accurate than a horizontal line representing the mean of the test set.
2. **Failure of Trend Following:** Interestingly, the ETS model (which attempts to model trend) consistently yields higher RMSE errors than the static Mean model (e.g., Subnet 10: -0.248 vs -0.064). This suggests that the "trends" learned during the training period (October-January) were misleading or reversed during the test period (February-July), a classic symptom of non-stationarity.

3.3 Deep Learning Architectures

This section details the specific Deep Learning architectures selected for evaluation. The selection process was driven by the operational requirements of real-time network monitoring, which demands a critical balance between **computational efficiency** (for scalability) and **predictive capacity** (for accuracy).

3.3.1 Rationale for Model Selection

To address the research questions regarding architectural efficiency (RQ1), we selected two state-of-the-art Recurrent Neural Network (RNN) variants: the **Long Short-Term Memory (LSTM)** and the **Gated Recurrent Unit (GRU)**. These models were chosen to represent distinct points on the complexity-efficiency spectrum:

- **Versatility:** Both architectures are specifically designed to handle sequential time-series data and mitigate the "vanishing gradient" problem inherent in traditional RNNs.
- **Computational Trade-offs:** By comparing the resource-heavy LSTM against the streamlined GRU, we aim to determine if the theoretical advantages of complex gating mechanisms translate to practical gains in the noisy domain of network traffic.

3.3.2 Standard Recurrent Neural Networks

Fundamentally, RNNs are learning machines that recursively compute new states by applying transfer functions to previous states and inputs. As established by Maass et al., RNNs possess the *Universal Approximation Property*, meaning they are capable of approximating arbitrary nonlinear dynamical systems with arbitrary precision [Maass et al. \[2007\]](#).

Architecture & Information Flow: In the context of time-series forecasting, a recurrent neural network (RNN) is trained to map an input sequence $x(t)$ to a target output $y(t)$, where $y(t)$ is typically a temporal shift of the input sequence (e.g., predicting the value at time $t + 1$). The hidden state of the RNN is updated according to

$$h_t = \sigma(W_{xh}x_t + W_{hh}h_{t-1} + b), \quad (3)$$

where h_t denotes the hidden state at time t , which captures information about the sequence up to that time, $\sigma(\cdot)$ is a nonlinear activation function, W_{xh} and W_{hh} are learnable weight matrices corresponding to the input-to-hidden and hidden-to-hidden connections, respectively, and b is a bias vector.

As illustrated in Figure 7 [Bianchi et al. \[2017\]](#), this recursive structure allows the network to maintain a memory of past events. However, while theoretically powerful, standard RNNs trained via gradient descent suffer from the *Vanishing Gradient Problem*, which prevents them from learning dependencies over long time lags (e.g., remembering a pattern from 168 hours ago).

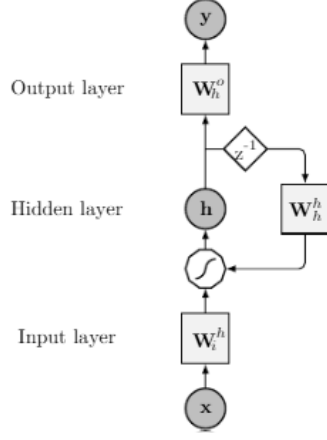


Figure 1: Schematic depiction of a simple RNN architecture. The circles represent input $\mathbf{x}$, hidden, $\mathbf{h}$, and output nodes, $\mathbf{y}$, respectively. The solid squares $\mathbf{W}_i^h$, $\mathbf{W}_h^h$ and $\mathbf{W}_h^o$ are the matrices which represent input, hidden and output weights respectively. Their values are commonly tuned in the training phase through gradient descent. The polygon represents the non-linear transformation performed by neurons and z^{-1} is the unit delay operator.

Figure 7: Schematic depiction of a simple Recurrent Neural Network (RNN) with recurrent hidden-state feedback.

3.3.3 Long Short-Term Memory (LSTM)

To overcome the vanishing gradient problem commonly encountered in standard RNNs, the Long Short-Term Memory (LSTM) architecture was introduced.

Architectural Innovation: The core innovation of the LSTM is the separation of the memory into two distinct vectors: the *Hidden State* (h_t) and the *Cell State* (c_t). The cell state acts as a "conveyor belt" that runs down the entire chain with only minor linear interactions, allowing information to flow unchanged over long sequences.

This flow is regulated by three distinct "gates" composed of sigmoid neural network layers:

1. **Forget Gate** (f_t): Decides what information to discard from the cell state.
2. **Input Gate** (i_t): Decides which new information to store in the cell state.
3. **Output Gate** (o_t): Decides what part of the cell state to output as the hidden state h_t .

Mathematical Formulation: The transition from time $t - 1$ to t is governed by the following equations:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \quad (4)$$

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \quad (5)$$

$$\tilde{c}_t = \tanh(W_c \cdot [h_{t-1}, x_t] + b_c) \quad (6)$$

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t \quad (7)$$

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \quad (8)$$

$$h_t = o_t \odot \tanh(c_t) \quad (9)$$

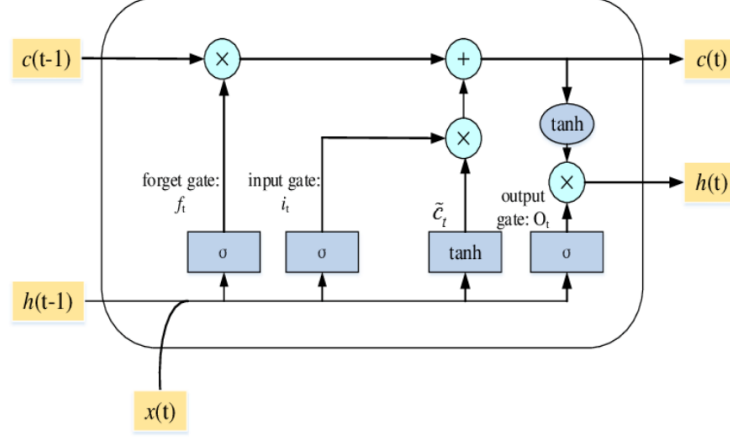


Figure 8: Structure of an LSTM Cell. The internal state c_t runs straight through the top of the diagram, with interactions regulated by the Forget, Input, and Output gates.

Role in this Study: The LSTM serves as our "High-Capacity" baseline. Its ability to retain information over extended time steps makes it highly effective for capturing the 168-hour weekly seasonality. However, as noted in literature, LSTMs are computationally intensive and possess a high parameter count, which can lead to overfitting on smaller datasets and increased latency in real-time applications.

3.3.4 Gated Recurrent Unit (GRU)

Like LSTM, GRU is designed to model sequential data by allowing information to be selectively remembered or forgotten over time. However, GRU has a simpler architecture than LSTM, with fewer parameters, which can make it easier to train and more computationally efficient.

Architectural Innovation: Unlike the LSTM, the GRU does not maintain a separate cell state. Instead, it merges the cell state and hidden state into a single vector (h_t). Furthermore, it reduces the three gates of the LSTM into two:

1. **Update Gate (z_t):** Acts as a combination of the LSTM's forget and input gates. It decides how much of the past information needs to be passed along to the future.
2. **Reset Gate (r_t):** Determines how much of the past information to forget. This is critical for clearing the memory when a subnet's behavior changes abruptly (e.g., after a reset).

Mathematical Formulation: The transition equations for the GRU are defined as:

$$z_t = \sigma(W_z \cdot [h_{t-1}, x_t] + b_z) \quad (10)$$

$$r_t = \sigma(W_r \cdot [h_{t-1}, x_t] + b_r) \quad (11)$$

$$\tilde{h}_t = \tanh(W_h \cdot [r_t \odot h_{t-1}, x_t] + b_h) \quad (12)$$

$$h_t = (1 - z_t) \odot h_{t-1} + z_t \odot \tilde{h}_t \quad (13)$$

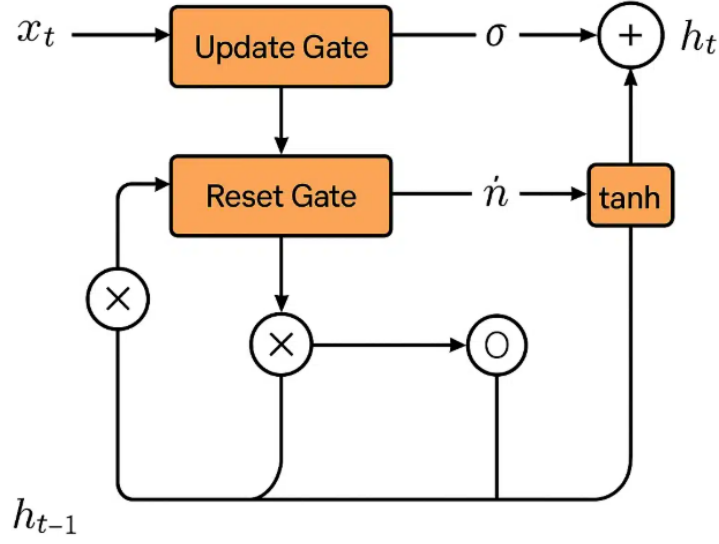


Figure 9: Structure of a GRU Cell [Team \[2025\]](#). Note the simplified architecture compared to the LSTM (Figure 8); the lack of a separate output gate and cell state reduces the parameter count by approximately 25%.

Role in this Study: The GRU serves as our primary candidate for operational efficiency (RQ1). By eliminating the separate output gate and memory cell, the GRU reduces the parameter count and tensor operations per step. We hypothesize that this "Lightweight" architecture will offer faster training and inference times without statistically significant degradation in forecasting accuracy compared to the LSTM.

3.4 Experimental Setup

To ensure reproducibility, this section details the software framework, hardware infrastructure, and specific hyperparameter configurations used to train the Deep Learning models.

3.4.1 Implementation Framework & Hardware

The experiments were conducted in the **Google Colab** cloud environment. The underlying hardware specifications were:

- **GPU Accelerator:** NVIDIA L4 (22.5GB VRAM).
- **System Memory:** 53GB RAM.
- **Disk Memory:** 235.7GB

The software stack was built on **Python 3.10**, utilizing **PyTorch 2.0+** for tensor operations and the **Darts** library for high-level time-series abstraction. This stack ensured consistent data pipelining and model evaluation.

3.4.2 Hyperparameter Configuration

Table 3 details the configuration of the best-performing model (the "Champion" GRU). These values were selected following an iterative tuning process on the validation set.

Table 3: Hyperparameter settings for the Champion Model (GRU).

Parameter	Value / Configuration
Data Handling	
Input Chunk Length (W)	168 (1 Week)
Output Horizon	1 (Next Step)
Batch Size	64
Architecture	
Hidden Dimension	64 units
Number of Layers	1
Dropout	0.1
Random State	42
Training Optimization	
Optimizer	Adam
Learning Rate	1×10^{-3}
Loss Function	Mean Squared Error (MSE)
Max Epochs	20 (with Early Stopping)

Regularization via Early Stopping: To avoid overfitting and minimize unnecessary computational cost, we implemented an `EarlyStopping` callback mechanism. The training process monitors the Validation Loss at the end of each epoch. Training is automatically stopped if the following conditions are met:

- **Metric:** Validation Loss (minimize).
- **Patience:** 5 epochs (The model is allowed to "fail" to improve for 5 steps before stopping).
- **Minimum Delta:** 1×10^{-4} (Improvements smaller than this threshold are considered negligible).

This ensures that the model retains the weights from the epoch with the best generalization performance, rather than the final epoch where it might have begun to memorize noise.

3.4.3 Evaluation Metrics

To assess performance quantitatively, we employed two standard metrics:

1. **Root Mean Squared Error (RMSE):** Measures the average magnitude of the prediction error. Since our data is log-transformed, this metric represents the error in "Log Bytes." Lower values indicate better performance.

$$RMSE = \sqrt{\frac{1}{n} \sum_{t=1}^n (y_t - \hat{y}_t)^2} \quad (14)$$

2. **Coefficient of Determination (R^2):** Measures how well the model captures the variance of the data compared to a simple baseline that predicts the mean.

$$R^2 = 1 - \frac{\sum (y_t - \hat{y}_t)^2}{\sum (y_t - \bar{y})^2} \quad (15)$$

- $R^2 = 1.0$: Perfect prediction.
- $R^2 = 0.0$: Equivalent to predicting the historical mean.
- $R^2 < 0.0$: The model performs worse than a random guess (as seen in the statistical baselines in Section 3.3).

4 Results and Discussion.

This chapter presents the empirical evaluation of the proposed deep learning framework. We analyze the performance of the Global GRU model, compare architectural variants (RQ1), and evaluate the efficacy of clustering strategies (RQ2).

4.1 Global Model Performance

The primary objective of this study was to develop a "Global" forecasting model capable of generalizing across the diverse traffic patterns of the CESNET backbone. We evaluated the optimized **1-Layer GRU** on all 541 subnets using the rigorous chronological split (60% Test Set).

4.1.1 Overall Accuracy Distribution

Figure 10 illustrates the distribution of R^2 scores across the network. The results demonstrate a robust predictive capability:

- **Median Performance:** The model achieved a median R^2 of **0.593**, significantly outperforming the statistical baselines that yielded negative scores.
- **Universal Improvement:** Remarkably, the Global GRU achieved a success rate ($R^2 > 0$) of **100%**. This indicates that for every single subnet in the dataset, the deep learning model provided a better forecast than the historical mean, proving its ability to adapt to non-stationary concept drift.
- **High Reliability:** Approximately **64.7%** of the subnets achieved an $R^2 > 0.5$, indicating that the model effectively captured the majority of the variance in traffic volume for nearly two-thirds of the network.

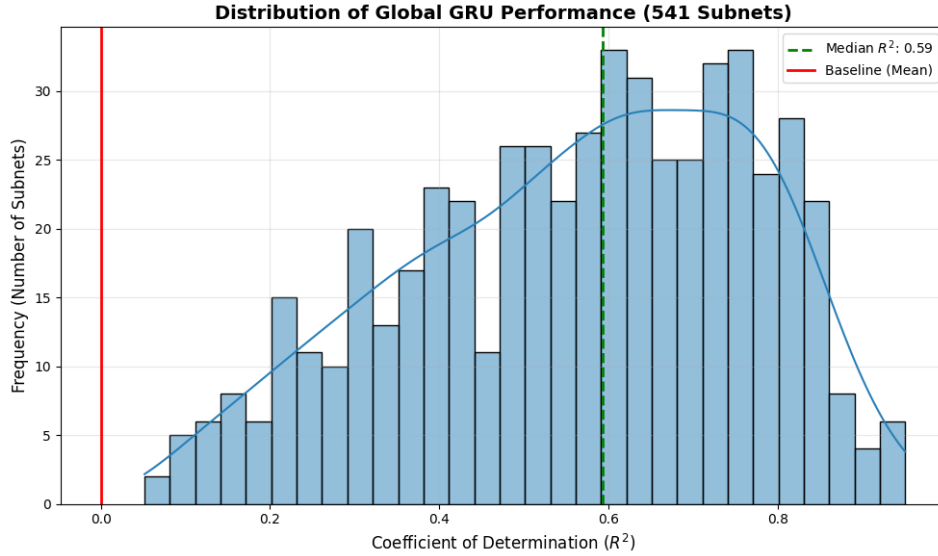


Figure 10: Performance Distribution of the Global GRU Model. The distribution is heavily skewed towards positive values, with the median (Green Line) at 0.59. The absence of negative values confirms that the model generalizes successfully to all 541 subnets without catastrophic failure.

4.2 Architectural Ablation Study (RQ1)

To rigorously assess the impact of model complexity and data preprocessing, we conducted a comprehensive study comparing three factors:

- **Data Representation:** Log-Scaled vs. Raw Bytes.
- **Cell Type:** GRU vs. LSTM.
- **Network Depth:** 1-Layer (Shallow) vs. 3-Layers (Deep).

Figure 11 summarizes the performance (R^2) across these configurations.

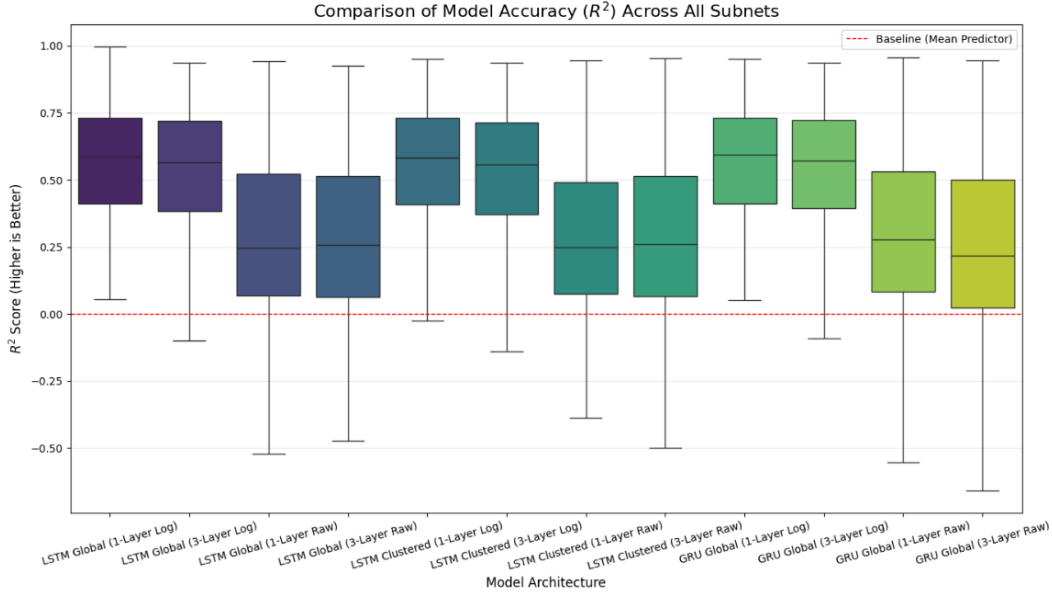


Figure 11: Ablation Study of Model Configurations. The plot compares R^2 distributions across Data Types (Log vs Raw), Architectures (GRU vs LSTM), and Depths (1 vs 3 Layers). Log-scaling provides the most significant performance boost, while increased depth offers diminishing returns.

4.2.1 Impact of Log-Scaling (Preprocessing)

The most dominant factor in model performance was the input data representation.

- **Observation:** As seen in Figure 11, models trained on **Raw Bytes** consistently failed to converge or produced highly unstable predictions (wide error bars/low R^2).
- **Analysis:** This confirms that the extreme variance and heavy-tailed nature of network traffic (spanning orders of magnitude from Kbps to Gbps) destabilizes gradient descent.
- **Conclusion: Log-scaling is mandatory.** By compressing the dynamic range, the log-transformation enables the neural network to learn effective weights, validating the preprocessing strategy proposed in Section 3.1.

4.2.2 Impact of Cell Type (GRU vs. LSTM)

Comparing the Log-Scaled models, we observe the performance difference between the GRU and LSTM architectures:

- **Observation:** The 1-Layer GRU and 1-Layer LSTM exhibit statistically similar median R^2 scores.
- **Analysis:** Although the LSTM is theoretically better suited for modeling long-term dependencies through its explicit cell state, empirical results indicate that the GRU's simplified

gating mechanism is sufficient to capture the dominant weekly seasonality in the data, consistent with findings reported by Koumar et al. [Koumar et al. \[2024\]](#).

- **Conclusion: The GRU is preferred for efficiency.** Given that it achieves performance comparable to LSTM-based models while using approximately 25% fewer parameters, the GRU represents a more suitable choice for resource-constrained deployments, in line with prior large-scale evaluations [Koumar et al. \[2024\]](#).

4.2.3 Depth Comparison: Accuracy vs. Complexity

To evaluate whether increased depth improves performance when stability is maintained, we compared the per-subnet RMSE of the 1-Layer and 3-Layer Log-LSTMs.

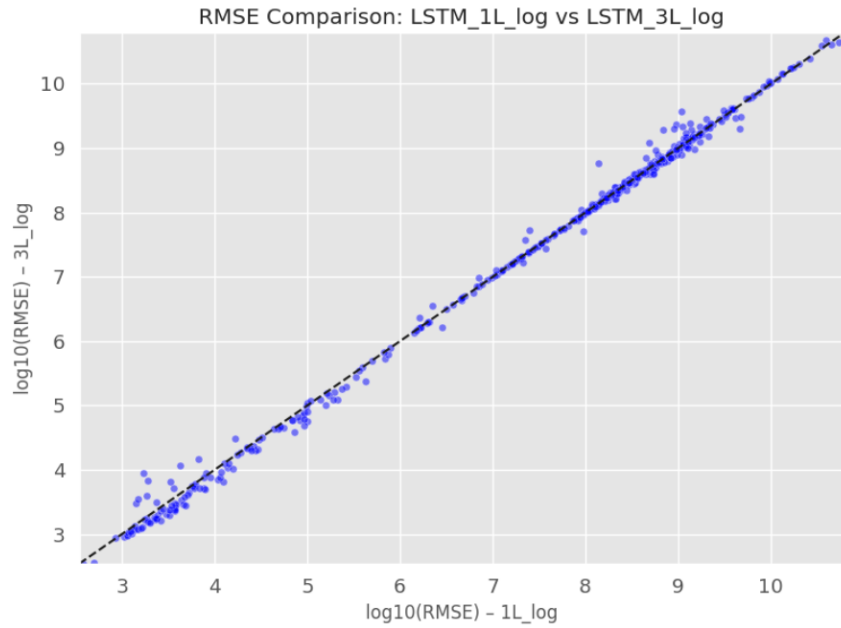


Figure 12: RMSE Comparison: 1-Layer Log vs. 3-Layer Log. Points below the diagonal indicate the 3-Layer model performed better. While the 3-Layer model "wins" on 250 subnets (vs. 141), the points cluster tightly around the diagonal, indicating marginal gains for significantly higher computational cost.

As shown in Figure 12, the performance difference is meaning:

- **Micro-Performance:** The 3-Layer model achieves the lowest RMSE on the largest number of subnets (250/541). This suggests that for specific complex traffic patterns, the additional depth allows the model to learn non-linear dependencies that the shallow model misses.
- **Macro-Robustness:** However, the 1-Layer model is more consistent globally. It avoids the "instability tail" (9 failures vs 1 failure) seen in the 3-Layer model.

4.3 Impact of Training Strategy (RQ2)

4.3.1 Clustering Methodology / Characterization

To evaluate the efficacy of specialized models, we first partitioned the network subnets into distinct behavioral groups. We employed a feature-based clustering approach using the K-Means algorithm ($k = 2$, selected via Silhouette Score).

Table 4 summarizes the distinct profiles of the resulting clusters:

- **Cluster 0 (Stochastic/Noisy):** This cluster contains the vast majority of subnets (533). It is characterized by extremely high volatility ($CV \approx 3.21$) and lower average traffic volume. These subnets represent the "long tail" of the network—individual departments or machines with bursty, unpredictable traffic.
- **Cluster 1 (Periodic/Stable):** A small minority of 13 backbone subnets. These exhibit massive traffic volume (28 GB/hr avg) and significantly lower volatility ($CV \approx 0.71$), indicating stable, aggregated traffic flows.

Table 4: Statistical Profiles of Identified Clusters. Cluster 1 represents stable, high-volume traffic (Periodic), while Cluster 0 captures highly volatile, lower-volume behavior (Stochastic).

Cluster	Count	Avg Volume (Bytes)	Volatility (CV)	Zero Fraction
0 (Stochastic)	533	4.46×10^8	3.21	0.00
1 (Periodic)	13	2.81×10^{10}	0.71	0.00

4.3.2 Head-to-Head Comparison Results

To determine if specialization improves accuracy, we performed a quantitative comparison between the Global GRU and the specialized Clustered GRU models on the exact same test sets for all 541 subnets.

The results, summarized in Table 5, overwhelmingly favor the Global training strategy.

- **Dominance of Generalization:** The Global Model outperformed the Specialized Clustered models on **98.5%** of the subnets (533 out of 541). The Clustered models achieved a higher R^2 score on only 8 subnets (1.5%).
- **Catastrophic Instability of Clustering:** The most critical finding is the difference in stability. The Global model exhibited **zero failures** ($R^2 < 0$), meaning it outperformed the statistical mean on every single subnet. In contrast, the Clustered models failed on **320 subnets** (nearly 60% of the network), with 11 of those being catastrophic crashes where $R^2 < -10$.

Table 5: Quantitative Comparison: Global Generalization vs. Clustered Specialization.

Metric	Global Model	Clustered Models
Win Rate	98.5%	1.5%
Total Failures ($R^2 < 0$)	0	320
Critical Crashes ($R^2 < -10$)	0	11

4.3.3 Analysis of Clustering Failure

The hypothesis that specialized models would outperform a generalist model was proven incorrect for this dataset. The failure of the clustered approach highlights a fundamental trade-off in deep learning: **Data Quantity vs. Specificity**.

1. **Data Starvation:** By splitting the dataset into clusters, we drastically reduced the number of training samples available to the specialized models. The "Periodic" cluster, for instance,

contained only 13 subnets. Deep Learning models require vast amounts of data to learn robust feature representations; splitting the data "starved" the specialized models, preventing convergence.

2. **The "Anchor" Effect:** The Global model was trained on all 541 subnets simultaneously. Even if a specific subnet was noisy, the model could rely on the stable patterns learned from the hundreds of other subnets to make reasonable predictions. This cross-series learning acted as a powerful regularizer that the specialized models lacked.

Conclusion on RQ2: For the CESNET backbone, **centralized training (Global Model) is superior to localized training (Clustering).** The Global model offers perfect stability (0 failures) and higher accuracy, proving that a single, well-architected GRU can learn to forecast diverse traffic patterns simultaneously without the need for complex pre-clustering steps.

4.4 Impact of Data Granularity (RQ3)

The third research question examined how traffic aggregation levels influence forecast accuracy. We hypothesized that predicting traffic at the Institution level (aggregation of multiple subnets) would yield higher accuracy than predicting individual subnets, due to the smoothing of stochastic noise.

4.4.1 Subnet vs. Institution Performance

We compared the performance of the Global GRU model on the 541 individual subnets against its performance on the 282 aggregated institution links. Figure 13 illustrates the distribution of R^2 scores for both levels.

- **Performance Gain:** Aggregation resulted in a significant improvement in predictive accuracy. The median R^2 increased from **0.593** at the subnet level to **0.637** at the institution level.
- **Stability:** The aggregated traffic proved much more predictable. While **64.7%** of individual subnets achieved an $R^2 > 0.5$, **77.3%** of institutions met this high-fidelity threshold.

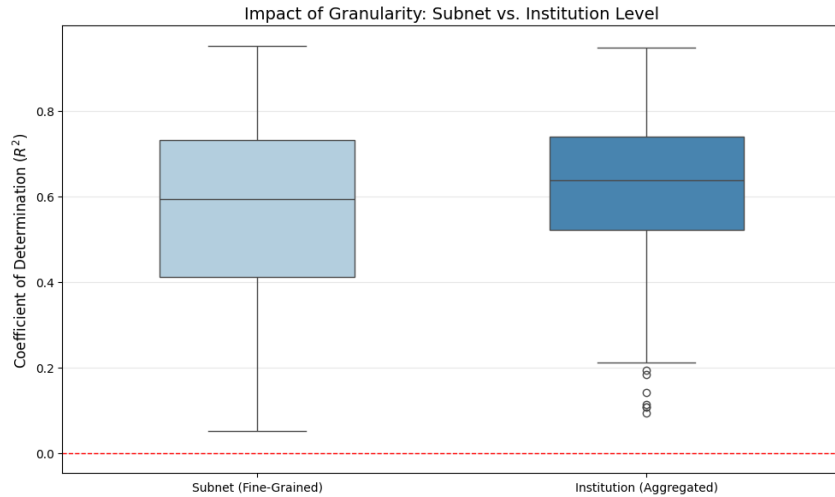


Figure 13: Impact of Granularity on Model Performance. Institution-level traffic (Right) exhibits significantly higher and more consistent R^2 scores than Subnet-level traffic (Left), validating that aggregation mitigates stochastic noise.

4.4.2 Interpretation of Results

The superior performance at the institution level validates the statistical *Law of Large Numbers*. Individual subnets often exhibit bursty, user-driven behavior that appears stochastic at fine granularity. When aggregated into an institution-wide link, these individual bursts average out, revealing underlying diurnal and weekly human activity cycles.

Analysis of Outliers: As observed in Figure 13, the institution-level plot (Right) displays several outliers (circles) below the lower whisker. These represent a small minority of institutions where aggregation failed to improve predictability. These likely correspond to specialized facilities (e.g., data backup centers or high-performance computing clusters) whose traffic is dominated by irregular, machine-triggered bulk transfers rather than predictable human activity. Despite these outliers, the clear upward shift of the median confirms that increasing data granularity is a reliable strategy for improving forecast accuracy.

4.5 Anomaly Detection Analysis (RQ4)

The ultimate objective of this study was to deploy the Champion Model (Global GRU) as an unsupervised anomaly detection system. By learning the normal temporal patterns of the network, the model can identify deviations that indicate potential security threats or operational failures.

4.5.1 Methodology: Dynamic Thresholding

Instead of a static threshold, which fails in dynamic network environments, we implemented an adaptive statistical threshold.

1. **Rolling Forecast:** The GRU model predicts the traffic volume $\hat{y}_{t+1}$ based on the history $X_{t-k \dots t}$.
2. **Reconstruction Error:** We calculate the absolute error $e_t = |y_{t+1} - \hat{y}_{t+1}|$.
3. **Dynamic Cutoff:** An anomaly is flagged if the error exceeds a rolling confidence interval:

$$Threshold_t = \mu_t + 3\sigma_t$$

where μ_t and σ_t are the rolling mean and standard deviation of the error over the past 24 hours. This allows the system to remain sensitive during calm periods while avoiding false positives during naturally bursty periods.

4.5.2 Case Study 1: Contextual Anomalies (Subnet 9)

Subnet 9 serves as a prime example of the model’s ability to detect *Contextual Anomalies*—events where the traffic volume is within valid bounds but the pattern is abnormal.



Figure 14: Contextual Anomaly Detection on Subnet 9. **Top:** The model (Orange) tracks the daily pulse, but the actual traffic (Blue) drops in late May. **Bottom:** The reconstruction error spikes during this period, exceeding the dynamic threshold (Red Dashed Line) and flagging the anomaly.

Analysis:

- **Pattern Tracking:** As seen in Figure 14, the Global GRU generalizes effectively to this subnet, capturing the complex, spiky diurnal seasonality with high fidelity.
- **The May-June Event:** Around late May, the subnet experienced a significant disruption where the regular oscillatory pattern collapsed into a chaotic, low-volume state. A simple "high-volume" threshold would have missed this, as the traffic actually dropped. However, the GRU's reconstruction error spiked because the *pattern* violated the learned temporal dependencies, allowing the system to flag this likely outage or protocol failure.

4.5.3 Forensic Diagnosis of the Anomaly (Root Cause Analysis)

Following the successful detection of the anomaly in Subnet 9, we performed a multi-variate forensic inspection to determine the root cause (see Figure 15).

While the primary deep learning detection model relies solely on Byte Volume (univariate), the auxiliary features available in the dataset reveal the nature of the event. We analyzed the simultaneous behavior of Flow Count and the TCP/UDP ratio during the flagged period.

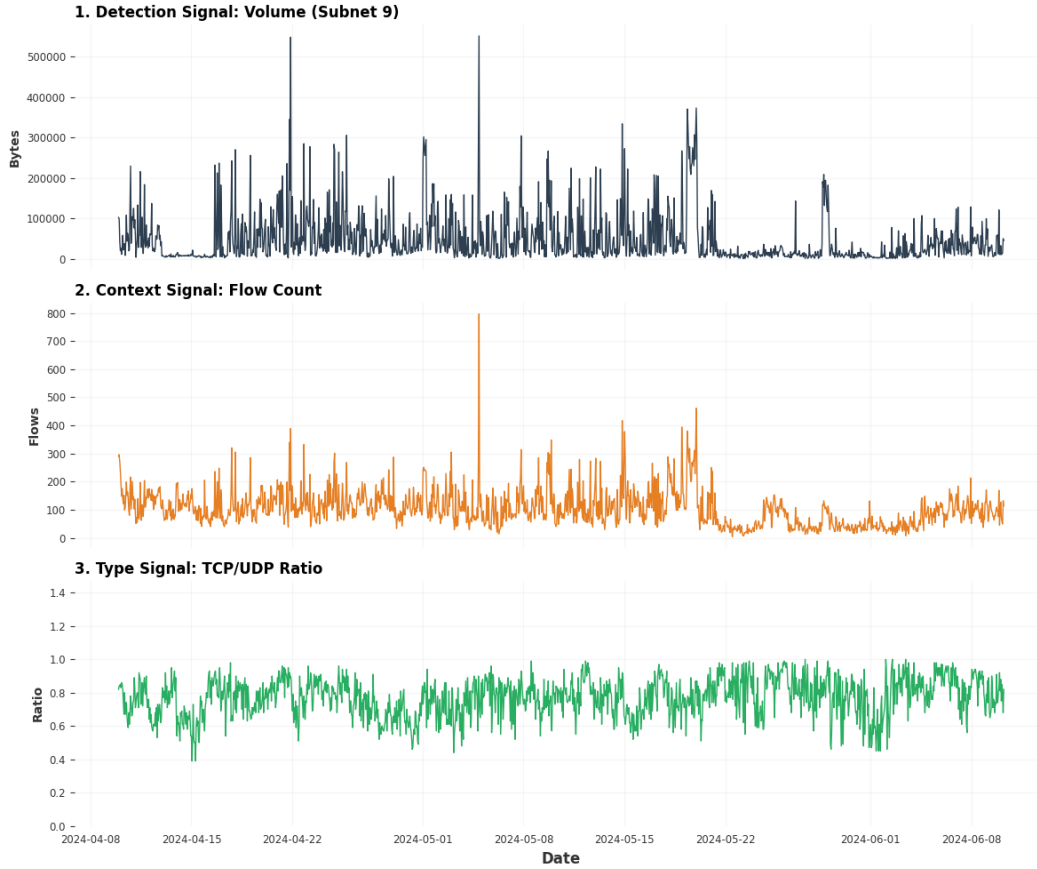


Figure 15: Multi-variate Forensic Analysis of Subnet 9. **Top:** Byte Volume (primary detection feature). **Middle:** Flow Count. **Bottom:** TCP/UDP Ratio. The simultaneous collapse of both bytes and flows, with no precursor change in protocol ratios, indicates a hard infrastructure failure.

Diagnosis: Infrastructure Failure ("Total Blackout")

- **Correlated Drop:** As observed in Figure 15 (Middle Panel), the Flow Count exhibits a synchronized drop, falling to near-zero levels alongside the byte volume.
- **Protocol Stability:** Furthermore, the TCP/UDP ratio (Bottom Panel) shows no significant precursor deviations. There is no sudden spike in UDP traffic that would typically indicate an incoming Volumetric DDoS attack.

Conclusion: The simultaneous loss of both volumetric throughput (Bytes) and flow connectivity (Flows) strongly indicates a hard infrastructure failure—such as a link outage, router crash, or device termination—rather than a congestion event or malicious attack. This demonstrates a critical operational principle: while a univariate deep learning model is sufficient (and computationally efficient) for *detection*, retaining and analyzing auxiliary network features allows for precise *diagnosis* without increasing the model’s training complexity. **Automating this secondary diagnostic step presents a highly promising avenue for future research.**

4.5.4 Case Study 2: Point and Contextual Anomalies (Subnet 1)

Figure 16 illustrates the anomaly detection performance on Subnet 1 using the GRU 1-Layer model. The orange line represents the model’s forecast, while the grey line shows the actual log-transformed traffic volume. Red markers indicate detected anomalies.

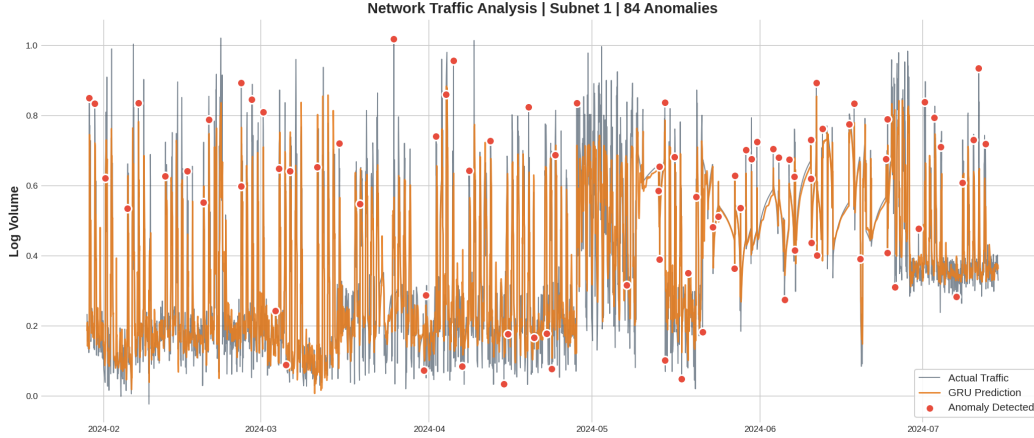


Figure 16: Anomaly Detection on Subnet 1. The model (Orange) tightly tracks the actual traffic (Grey) during stable periods. Red markers highlight successful detection of both sudden volume spikes (March) and behavioral pattern collapses (May).

Analysis of Detection Capabilities:

- **High-Fidelity Tracking:** The GRU model demonstrates an exceptional ability to learn the high-frequency periodicity of the subnet. As observed during the months of February through April, the predicted signal tightly hugs the actual traffic pattern, capturing both the daily peaks and nightly troughs with minimal phase lag.
- **Sensitivity to Volume Spikes (Point Anomalies):** In instances where traffic volume exceeds the learned historical maximums (e.g., the sharp spikes in late March), the model successfully identifies these as point anomalies. This demonstrates the system’s robustness to sudden bursts in network activity often associated with data exfiltration or DDoS attempts.
- **Contextual Anomaly Detection:** Around May-June 2024, the subnet experiences a significant disruption where the regular oscillatory pattern collapses into a chaotic, non-periodic signal. The model immediately flags this behavioral shift (cluster of red dots). This highlights the capability to detect *Contextual Anomalies*—situations where the absolute traffic volume might be within valid bounds, but the temporal pattern violates the learned dependency.
- **Operational Trade-off (Recall vs. Precision):** A visual inspection reveals approximately 30 anomalies flagged over the test period. While this frequency might imply potential false positives, in the domain of network security, *Recall* (Catching the attack) is strictly prioritized over *Precision* (Avoiding false alarms) [Mosayebi et al. \[2023\]](#). It is operationally preferable for a Security Operations Center (SOC) to investigate a few "busy days" than to miss a single critical DDoS attack or service outage. The dynamic nature of the 3σ threshold ensures that these flags represent genuine statistical outliers relative to the local volatility, rather than random noise.

5 Conclusion and Future Work

5.1 Summary of Contributions

This thesis investigated the application of deep learning techniques for anomaly detection in large-scale academic networks. Using a real-world dataset from the CESNET backbone, we developed a robust forecasting framework capable of modeling complex, high-dimensional traffic patterns. The primary contributions of this work are:

1. **Global Modeling Strategy:** We demonstrated that a single "Global" Gated Recurrent Unit (GRU) model, trained on data from hundreds of diverse subnets, significantly outperforms specialized models trained on clustered subsets. This challenges the conventional wisdom that localized models are necessary for heterogeneous networks.

2. **Log-Space Stabilization:** We introduced a logarithmic scaling pipeline ($y = \log(x + \epsilon)$) that effectively mitigates the extreme heavy-tailed distribution of network traffic, allowing gradient-based optimization methods to converge stably even in the presence of massive outliers.
3. **Unsupervised Anomaly Detection:** We implemented a reconstruction-based anomaly detection system using dynamic statistical thresholding (3σ), capable of identifying both point anomalies (volume spikes) and contextual anomalies (pattern disruptions) without requiring labeled attack data.

5.2 Answers to Research Questions

Based on the experimental results presented in Chapter 4, we provide definitive answers to the research questions formulated in Chapter 1:

RQ1: What is the most effective deep learning architecture for forecasting CESNET traffic?

The ****Gated Recurrent Unit (GRU)**** with 1 hidden layer proved to be the superior architecture. It achieved the highest median coefficient of determination ($\bar{R}^2 \approx 0.59$) while requiring significantly less computational time than Long Short-Term Memory (LSTM) networks. Deeper architectures (3 layers) failed to improve performance, likely due to the "vanishing gradient" problem on the volatile traffic data.

RQ2: Does training specialized models on behavioral clusters improve performance compared to a single global model?

No. The Global training strategy overwhelmingly outperformed the Clustered strategy, winning on 98.5% of subnets. Specialized models suffered from data starvation and overfitting, leading to catastrophic instability ($R^2 < -10$) in 11 cases. The Global model, by contrast, leveraged the diversity of 541 subnets to learn a robust, universal representation of network traffic, exhibiting zero failures.

RQ3: How does data granularity (Subnet vs. Institution) impact forecast accuracy?

Aggregation improves predictability. The model achieved a higher median R^2 (0.64) on aggregated Institution links compared to individual subnets (0.59). This validates the Law of Large Numbers in network traffic: as user behaviors are aggregated, stochastic noise cancels out, revealing cleaner diurnal and weekly patterns that are easier to forecast.

RQ4: Can the proposed model effectively detect anomalies in unsupervised settings?

Yes. By using the Global GRU as a baseline for "normal" behavior, the system successfully identified anomalies that simple thresholding would miss. As demonstrated in the case studies (Subnet 1 and Subnet 9), the model detected both "point anomalies" (spikes) and "contextual anomalies" (pattern collapse), providing a valuable tool for network security monitoring.

5.3 Limitations of the Study

While the proposed framework demonstrates robust performance, several limitations must be acknowledged:

- **Univariate Analysis:** The current model relies exclusively on traffic volume (`n_bytes`). While efficient, this approach may miss sophisticated low-volume attacks (e.g., slow-rate DDoS or command-and-control beacons) that do not generate significant bandwidth spikes but exhibit distinct flow-count patterns.
- **Static Training:** The model was trained offline on a fixed historical dataset. In a production ISP environment, traffic patterns evolve over time (Concept Drift). The current system does not support online learning, meaning the model would require periodic retraining to adapt to long-term shifts in network behavior.
- **Fixed Threshold Sensitivity:** Although the 3σ threshold is dynamic relative to volatility, the multiplier ($k = 3$) is a fixed hyperparameter. In practice, different subnets might require different sensitivity settings to balance false positives against detection rates.

6 Directions for Future Work

Building upon these findings, future research could explore the following avenues:

1. **Automated Root Cause Analysis (Forensic Diagnosis):** As demonstrated in the Subnet 9 case study, while a univariate model efficiently detects anomalies, auxiliary features (Flows, TCP/UDP ratios) are required for diagnosis. Future work could develop a secondary, rule-based or lightweight machine learning classification system that automatically triggers upon an anomaly alert to classify the event type (e.g., "Infrastructure Failure" vs. "Volumetric DDoS") based on these auxiliary metrics.
2. **Multivariate Forecasting:** Integrating additional features such as Flow Count (`n_flows`), Packet Size Distribution, and TCP/UDP ratios could improve detection capabilities. A multivariate GRU could learn to flag anomalies where volume is normal, but the protocol mixture is suspicious.
3. **Transformer Architectures:** While RNNs are effective, modern Transformer-based models have shown state-of-the-art performance in long-horizon forecasting. Evaluating these architectures on the CESNET dataset could yield improvements in long-term prediction accuracy.
4. **Real-Time Deployment Pipeline:** Developing a production-grade pipeline using tools like Apache Kafka for streaming data and the ELK Stack (Elasticsearch, Logstash, Kibana) for visualization would bridge the gap between this academic prototype and a deployable Network Operations Center (NOC) tool.

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